THE TRACE OF AN ENDOMORPHISM OF A PERFECT COMPLEX IS WELL DEFINED IN THE COMMUTATOR QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file trace.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Every morphism in the derived category between such complexes is represented by an actual map of complexes, well defined up to homotopy (derived-lemma-morphisms-from-projective-complex). invoked at proof lines 1055-1057 and used throughout. Provides the representatives f^bullet, gamma_1^bullet, gamma_2^bullet the computation acts on.

Source reference. Original lemma 064B: sources/stacks-chow/tags.txt, source lines 7860–7860. Announced at proof lines 1055-1057 and used throughout to represent f, gamma_1, gamma_2 by maps of complexes; load-bearing inside the counted span.

Proposition 0.2. *Let $K \in D_{perf}(\Lambda)$ and $f \in End_{D(\Lambda)}(K)$. Then the trace $Tr(f) \in \Lambda^\natural$ is well defined.*

Proof. We will use Derived Categories, Lemma 064B without further mention in this proof. Let $P^\bullet$ be a bounded complex of finite projective Λ-modules and let $\alpha : P^\bullet \to K$ be an isomorphism in $D(\Lambda)$. Then $\alpha^{-1} \circ f \circ \alpha$ corresponds to a morphism of complexes $f^\bullet : P^\bullet \to P^\bullet$ well defined up to homotopy. Set

$$\mathrm{Tr}(f) = \sum_i (-1)^i \mathrm{Tr}(f^i : P^i \to P^i) \in \Lambda^\natural.$$

Given $P^\bullet$ and α, this is independent of the choice of $f^\bullet$. Namely, any other choice is of the form $\tilde{f}^\bullet = f^\bullet + dh + hd$ for some $h^i : P^i \to P^{i-1} (i \in \mathbf{Z})$. But

$$\begin{aligned}
\mathrm{Tr}(dh) &= \sum_i (-1)^i \mathrm{Tr}(P^i \xrightarrow{dh} P^i) \\
&= \sum_i (-1)^i \mathrm{Tr}(P^{i-1} \xrightarrow{hd} P^{i-1}) \\
&= -\sum_i (-1)^{i-1} \mathrm{Tr}(P^{i-1} \xrightarrow{hd} P^{i-1}) \\
&= -\mathrm{Tr}(hd)
\end{aligned}$$

Date: stacks-project:trace.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

and so $\sum_i (-1)^i \mathrm{Tr}((dh+hd)|_{P^i}) = 0$. Furthermore, this is independent of the choice of $(P^\bullet, \alpha)$: suppose $(Q^\bullet, \beta)$ is another choice. The compositions

$$Q^\bullet \xrightarrow{\beta} K \xrightarrow{\alpha^{-1}} P^\bullet \quad \text{and} \quad P^\bullet \xrightarrow{\alpha} K \xrightarrow{\beta^{-1}} Q^\bullet$$

are representable by morphisms of complexes $\gamma_1^\bullet$ and $\gamma_2^\bullet$ respectively, such that $\gamma_1^\bullet \circ \gamma_2^\bullet$ is homotopic to the identity. Thus, the morphism of complexes $\gamma_2^\bullet \circ f^\bullet \circ \gamma_1^\bullet : Q^\bullet \to Q^\bullet$ represents the morphism $\beta^{-1} \circ f \circ \beta$ in $D(\Lambda)$. Now

$$\begin{aligned} \mathrm{Tr}(\gamma_2^\bullet \circ f^\bullet \circ \gamma_1^\bullet|_{Q^\bullet}) &= \mathrm{Tr}(\gamma_1^\bullet \circ \gamma_2^\bullet \circ f^\bullet|_{P^\bullet}) \\ &= \mathrm{Tr}(f^\bullet|_{P^\bullet}) \end{aligned}$$

by the fact that $\gamma_1^\bullet \circ \gamma_2^\bullet$ is homotopic to the identity and the independence of the choice of $f^\bullet$ we saw above. $\square$